Lesson 5.6 - Probability of Mutually Exclusive Events with Venn Diagrams

> **Video:** Probability of Mutually Exclusive Events With Venn Diagrams by The Organic Chemistry Tutor
> **Chapter:** 5 - Introduction to Probability

Definition

Events A and B are **mutually exclusive** if they **cannot happen at the same time**, meaning they share **no outcomes**:

$$A \cap B = \varnothing \quad \iff \quad P(A \text{ and } B) = 0$$

Venn diagram: mutually exclusive events are drawn as **separate, non-overlapping** circles. Events that are not mutually exclusive are drawn as **overlapping** circles.

Testing for mutual exclusivity

One die, $S = \{1, 2, 3, 4, 5, 6\}$:

- $A = \{1, 2, 3\},\ B = \{5, 6\},\ C = \{3, 4, 5\}$

Pair	Intersection	$P(\text{both})$	Mutually exclusive?
A, B	$\varnothing$	0	**Yes**
A, C	{3}	1/6	No
B, C	{5}	1/6	No

> **Rule:** if $P(A \text{ and } B) = 0$, they are mutually exclusive. If it is anything other than 0, they are not.

The addition rule

General (always works):

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Mutually exclusive (the last term is 0):

$$P(A \text{ or } B) = P(A) + P(B)$$

Example 1 - overlapping events

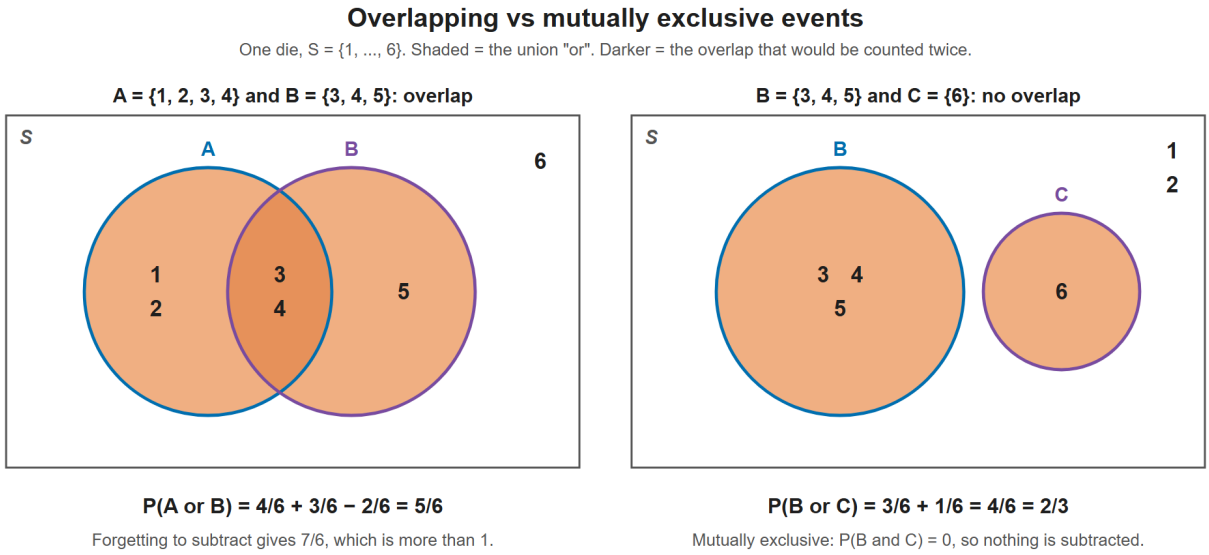
$A = \{1, 2, 3, 4\},\ B = \{3, 4, 5\}$. They share 3 and 4, so they are **not** mutually exclusive.

Venn diagram regions: A only = {1, 2}, both = {3, 4}, B only = {5}, outside both = {6}.

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) = \frac{4}{6} + \frac{3}{6} - \frac{2}{6} = \frac{5}{6}$$

Without subtracting $\frac{2}{6}$ you would get $\frac{7}{6}$, which is more than 1 and therefore impossible. The subtraction removes the double-counted 3 and 4.

Check: $A \cup B = \{1, 2, 3, 4, 5\}$, which is 5 of 6 outcomes ✓



Overlapping circles share outcomes, so the overlap must be subtracted once. Separate circles share nothing, so P(A or B) is simply P(A) + P(B).

Example 2 - mutually exclusive events

$B = \{3, 4, 5\},\ C = \{6\}$. They share nothing, so they are mutually exclusive and the circles don't overlap.

$$P(B \text{ or } C) = P(B) + P(C) - P(B \text{ and } C) = \frac{3}{6} + \frac{1}{6} - 0 = \frac{4}{6} = \frac{2}{3}$$

Key takeaways

- Mutually exclusive = no shared outcomes = $P(A \cap B) = 0$ = non-overlapping circles.
- Use $P(A) + P(B) - P(A \cap B)$ in general. For mutually exclusive events it simplifies to $P(A) + P(B)$.
- A probability above 1 is a sign that an overlap was counted twice.